

STAT452: Multiple Linear Regression — Lesson 1 Notes

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1 Introduction to Statistical and Linear Models

1.1 Conceptual Overview

- **Statistical Model:** A simple, low-dimensional summary designed to capture data relationships, the data-generation process, and the underlying relationships present in a population while explicitly accounting for real-world uncertainty or noise.
- **Linearity Definition:** A regression model is structurally defined as “linear” if it is linear in its **parameters** $(\beta_0, \beta_1, \dots, \beta_p)$, regardless of any non-linear transformations applied to the predictor variables (e.g., X^2 , $\log(X)$, or reciprocal transformations).
- **MLR vs. SLR:** While Simple Linear Regression (SLR) handles exactly one predictor variable X , Multiple Linear Regression (MLR) integrates multiple predictor variables $(X_1, X_2, \dots, X_p)$ to systematically model a single response variable Y .

1.2 Model Specifications

- **Scalar Form:** For an individual observation i (where $i = 1, 2, \dots, n$), the relationship is modeled as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

where ϵ_i represents the random error or noise terms, assumed to be independent and identically distributed (*i.i.d.*) following a normal distribution:

$$\epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- **Matrix Form:** To represent all n cases simultaneously, the system is compactly expressed as:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where:

$$\mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}_{n \times (p+1)}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}_{(p+1) \times 1}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}_{n \times 1}$$

1.3 Population vs. Sample Regression Lines

- **Population Regression Line ($Y = \mathbf{X}\boldsymbol{\beta}$):** Represents the true, fixed, but unknown relationship governing the entire population. It is the primary object of statistical interest.
- **Sample Regression Line ($Y = \mathbf{X}\hat{\boldsymbol{\beta}}$):** Calculated directly from sample data. It is inherently random and varies from sample to sample.

2 Least Squares Estimation

2.1 Objective and Derivation

The goal of Least Squares (LS) estimation is to find parameter estimates $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ that minimize the Least Squares criterion function L , which aggregates the sum of squared vertical distances from the data points to the fitted hyperplane:

$$L(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p) = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2 = \boldsymbol{\epsilon}^T \boldsymbol{\epsilon} = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

Taking partial derivatives with respect to each $\hat{\beta}_j$ and setting them to 0 yields a system of $(p+1)$ equations known as the **Normal Equations**:

$$\frac{\partial L}{\partial \hat{\beta}_0} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip}) = 0$$

$$\frac{\partial L}{\partial \hat{\beta}_k} = -2 \sum_{i=1}^n x_{ik} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip}) = 0, \quad k = 1, 2, \dots, p$$

2.2 Core Least Squares Formulas

- **Matrix Normal Equation:**

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y}$$

- **Least Squares Estimator ($\hat{\boldsymbol{\beta}}$):** Assuming $\mathbf{X}^T \mathbf{X}$ is non-singular and invertible, the unique solution is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

- **Unbiased Property:** Assuming the design matrix $\mathbf{X}$ is fixed, the LS estimator is strictly unbiased:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$$

3 Fitted Model and Residual Properties

3.1 Fitted Values and Residuals

- **Fitted (Predicted) Value:** The estimate of the response variable generated by the fitted model:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}$$

- **Residual (e_i):** The difference between the actual observed value and the estimated fitted value:

$$e_i = y_i - \hat{y}_i$$

Note that $e_i \neq \epsilon_i$ because $\hat{\beta}_j \neq \beta_j$ due to sampling variation.

3.2 Mathematical Properties of Residuals

- **Zero Sum Condition:** Residuals always sum to zero:

$$\sum_{i=1}^n e_i = 0$$

- **Orthogonality to Predictors:** Residuals are perfectly orthogonal to all explanatory variables:

$$\sum_{i=1}^n x_{ik} e_i = 0, \quad \text{for } k = 1, 2, \dots, p$$

- **Zero Correlation:** The combined properties above mathematically guarantee that the residuals maintain a 0 correlation with each of the p predictors:

$$\text{Cov}(X_k, e) = \frac{1}{n-1} \left(\sum_{i=1}^n x_{ik} e_i - n \bar{x}_k \bar{e} \right) = 0$$

3.3 Error Variance Estimation

- **Mean Square Error (MSE):** The true error variance σ^2 is unknown and must be estimated via the Mean Square Error (MSE). It divides the residual sum of squares by the error degrees of freedom ($n - p - 1$) to provide an unbiased estimator:

$$\sigma^2 \approx \text{MSE} = \frac{\sum_{i=1}^n e_i^2}{n - p - 1}$$

4 Inference and Hypothesis Testing for Coefficients

4.1 Variance-Covariance Structure

The variance-covariance matrix of the parameter estimator vector $\hat{\beta}$ in matrix notation is structured as:

$$\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$$

$$\text{Var}(\hat{\beta}) = \begin{bmatrix} \text{Var}(\hat{\beta}_0) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) & \dots & \text{Cov}(\hat{\beta}_0, \hat{\beta}_p) \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_0) & \text{Var}(\hat{\beta}_1) & \dots & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\beta}_p, \hat{\beta}_0) & \text{Cov}(\hat{\beta}_p, \hat{\beta}_1) & \dots & \text{Var}(\hat{\beta}_p) \end{bmatrix}$$

- **Standard Error (SE):** Calculated by taking the square root of the corresponding diagonal elements of $(\mathbf{X}^T \mathbf{X})^{-1}$ and substituting the population parameter σ^2 with the sample MSE:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\text{MSE} \times [(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}$$

4.2 Hypothesis Testing (*t*-test)

To test the significance of an isolated individual coefficient ($H_0 : \beta_j = \beta_j^0$ vs. $H_1 : \beta_j \neq \beta_j^0$):

- **Test Statistic:**

$$t = \frac{\hat{\beta}_j - \beta_j^0}{\text{SE}(\hat{\beta}_j)}$$

which strictly follows a Student's *t*-distribution with degrees of freedom $df = n - p - 1$ under H_0 .

- **Two-Tailed *P*-value Calculation (via R):**

$$\text{P-value} = 2 \times \text{pt}(\text{abs}(t), \text{df} = n - p - 1, \text{lower.tail} = \text{FALSE})$$

4.3 Confidence Intervals

The $100(1 - \alpha)\%$ confidence interval for any regression parameter β_j is evaluated as:

$$\hat{\beta}_j \pm t_{(n-p-1, \alpha/2)} \times \text{SE}(\hat{\beta}_j)$$

where $t_{(n-p-1, \alpha/2)}$ represents the critical value of the t_{n-p-1} distribution at a significance level of α .

5 Goodness-of-Fit Metrics

- **Multiple R^2 (Coefficient of Determination):** Represents the total proportion of systemic variability in Y successfully explained by the set of predictors $X_1, \dots, X_p$:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$$

Where $\text{SST} = \sum (y_i - \bar{y})^2$, $\text{SSE} = \sum (y_i - \hat{y}_i)^2$, and $\text{SSR} = \sum (\hat{y}_i - \bar{y})^2$. Note that R^2 will always increase or remain static as more variables are introduced.

- **Adjusted R^2 (R_{adj}^2):** Modifies R^2 by penalizing the model for adding unnecessary predictors based on degrees of freedom constraints:

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/df_E}{\text{SST}/df_T} = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)} = 1 - \frac{n - 1}{n - p - 1} (1 - R^2)$$

Unlike Multiple R^2 , R_{adj}^2 can decrease if non-essential features are integrated.

6 Model Comparison and Nested Testing

6.1 Conceptual Framework

Model 1 is structurally defined as **nested** within Model 2 if Model 1 is a restricted special case of Model 2 (created by constraining parameters, e.g., setting specific $\beta_k = 0$).

- The simpler model is termed the **reduced model**.
- The complex model is termed the **full model**.
- Models using identical datasets share a completely identical Total Sum of Squares (SST).
- Because the full model searches a broader parameter space, its optimization bounds guarantee:

$$\text{SSE}_{\text{reduced}} \geq \text{SSE}_{\text{full}} \quad \text{and} \quad \text{SSR}_{\text{reduced}} \leq \text{SSR}_{\text{full}}$$

6.2 The Partial F -Test Formula

To test H_0 : Reduced model is true vs. H_1 : Full model is true:

$$F = \frac{(\text{SSE}_{\text{reduced}} - \text{SSE}_{\text{full}}) / (df_E^{\text{reduced}} - df_E^{\text{full}})}{\text{MSE}_{\text{full}}}$$

Under H_0 , this statistic follows an F -distribution with degrees of freedom:

$$df_1 = df_E^{\text{reduced}} - df_E^{\text{full}}, \quad df_2 = df_E^{\text{full}}$$

6.3 Overall Model Significance Test

When testing whether *any* of the selected predictors are useful ($H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$), the reduced model simplifies to an intercept-only model ($y_i = \beta_0 + \epsilon_i$), where $\text{SSE}_{\text{reduced}} = \text{SST}_{\text{full}}$ and $df_E^{\text{reduced}} = n - 1$. The partial F -test collapses to the overall model F -test:

$$F = \frac{(\text{SST}_{\text{full}} - \text{SSE}_{\text{full}}) / [n - 1 - (n - p - 1)]}{\text{SSE}_{\text{full}} / (n - p - 1)} = \frac{\text{SSR}_{\text{full}} / p}{\text{SSE}_{\text{full}} / (n - p - 1)} = \frac{\text{MSR}_{\text{full}}}{\text{MSE}_{\text{full}}}$$

which follows an $F_{p, n-p-1}$ distribution under H_0 .

7 Residual Analysis (Model Diagnostics)

Residual analysis is performed to evaluate the four core Gauss-Markov and normality assumptions of linear regression models:

Assumption	Mathematical Definition	Diagnostic Check Method
1. Linearity	$\mathbb{E}[Y \mathbf{X} = \mathbf{x}] = \mathbf{x}^T \boldsymbol{\beta}$	Residual plot: e_i vs. $\hat{y}_i$; curves indicate non-linearity.
2. Homoscedasticity	$\text{Var}(Y \mathbf{X} = \mathbf{x}) = \sigma^2$	Residual plot; funnel or megaphone shapes show non-constant variance.
3. Independence	$\text{Cov}(\epsilon_i, \epsilon_j) = 0 \quad \forall i \neq j$	Residual plot ordered sequentially.
4. Normality	$Y \mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathbf{x}^T \boldsymbol{\beta}, \sigma^2)$	Normal Q-Q plot; points should lie near the diagonal reference line.

8 Casio fx-580VN X Exam Utility Sections

If a Casio fx-580VN X scientific calculator is authorized during examination, specific arithmetic subsets can be automated rapidly to verify raw matrix calculations and minimize algebraic errors.

8.1 Automating Matrix Operations for Coefficient Estimation

When the examination question provides a small, explicit data matrix ($\mathbf{X}$ and $\mathbf{Y}$) where n is small and $p \leq 3$, you can bypass structural manual matrix arithmetic via the calculator's dedicated **Matrix Mode** (MENU $\rightarrow$ 4).

1. Data Input Matrix Assignments:

- Define Matrix A (MatA) as the design matrix $\mathbf{X}$ with dimension $n \times (p + 1)$.
- Define Matrix B (MatB) as the column response vector $\mathbf{Y}$ with dimension $n \times 1$.

2. Generating Summary Matrices Directly:

- To obtain the matrix component $\mathbf{X}^T \mathbf{X}$, execute the command block:

$$\text{MatA}^T \times \text{MatA}$$

(Using Trn in the OPTN menu to compute the transposition).

- To obtain the column matrix component $\mathbf{X}^T \mathbf{Y}$, execute the command block:

$$\text{MatA}^T \times \text{MatB}$$

- #### 3. Direct Parameter Verification ($\hat{\beta}$):
- If the computed $\mathbf{X}^T \mathbf{X}$ yields a square matrix of size $\leq 4 \times 4$, assign this resulting matrix to a new matrix slot, say Matrix C (MatC). The final least squares coefficient parameters can be verified instantly by computing the inverse matrix multiplication chain:

$$\hat{\beta} = \text{MatC}^{-1} \times (\text{MatA}^T \times \text{MatB})$$

8.2 Solving System of Normal Equations via Simultaneous Equation Solvers

When the problem provides the fully evaluated scalar expansions of the Normal Equations rather than a data table, calculation speed can be maximized using the **Equation/Func Mode** (MENU $\rightarrow$ 9 $\rightarrow$ 1).

- Select the number of unknowns equal to $p + 1$ (the total parameters $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$).
- Directly input the consolidated summation parameters:

$$\left[\begin{array}{cccc|c} n & \sum x_{i1} & \dots & \sum x_{ip} & \sum y_i \\ \sum x_{i1} & \sum x_{i1}^2 & \dots & \sum x_{i1}x_{ip} & \sum x_{i1}y_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \sum x_{ip} & \sum x_{ip}x_{i1} & \dots & \sum x_{ip}^2 & \sum x_{ip}y_i \end{array} \right]$$

- Pressing = will output the optimized exact fractions or decimals for $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ concurrently, serving as an absolute confirmation for manually computed entries.

8.3 Two-Variable Summary Statistics Automation (SLR Check)

For sub-problems restricting data checking to a Simple Linear Regression framework or analyzing interactions between two structural attributes X_k and Y , use the **Statistics Mode** (MENU $\rightarrow$ 6 $\rightarrow$ 2 for $y = a + bx$).

- Input data pairs directly into the x and y spreadsheets.

- Pressing **OPTN** → **3** (2-Variable Regression Calc) outputs structural metrics immediately:
 - a : The sample intercept ($\hat{\beta}_0$).
 - b : The sample slope ($\hat{\beta}_1$).
 - r : The Pearson correlation coefficient (r_{xy}). Squaring this manually gives the Multiple R^2 value for the simple model ($R^2 = r_{xy}^2$).
- Pressing **OPTN** → **2** (2-Variable Variable Calc) retrieves fundamental summation quantities ($\sum x, \sum x^2, \sum y, \sum y^2, \sum xy$) without needing manual step-by-step arithmetic loops.

8.4 Validating Unconditional Sum of Squares via 1-Variable Summary Statistics

To quickly verify the denominator for variance allocations without manual deviation squaring ($SST = \sum (y_i - \bar{y})^2$):

- Enter the raw Y data entries into **Statistics Mode: 1-Variable** (**MENU** → **6** → **1**).
- Execute **OPTN** → **2** (1-Variable Calc).
- Extract the value of $\sum x^2$ (which acts as $\sum y_i^2$) and $\sum x$ (which acts as $\sum y_i$).
- Apply the computational shortcut formula to verify your manual total sum of squares calculation:

$$SST = \sum y_i^2 - \frac{(\sum y_i)^2}{n}$$